

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Environmental Engineering) SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Environmental Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTCESS801E - PLASTIC WASTE MANAGEMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The resin code of Low density polyethylene (LDPE) is ____
a) 1 b) 3 c) 4 d) 6
- 2 The country which produces largest plastic material is ____
a) India b) USA c) China d) Russia
- 3 _____ are the small pieces of plastic derived from the breakdown of larger plastic debris.
a) Micro beads b) Micro plastics c) Plastic pellets d) Scrubbers
- 4 The largest producer of plastic waste in India according to CPCB report, 2017 is ____
a) Tamilnadu b) Maharashtra c) Gujarat d) Punjab
- 5 Ensuring that open burning of plastic waste does not take place' is the responsibility of ____
a) Local bodies b) Gram Panchayats c) Producers d) Both (a) & (b)
- 6 Minimum thickness of plastic carry bags allowed, according to Plastic waste management rules (2016) is ____ microns.
a) 30 b) 40 c) 50 d) 60
- 7 The first state in India to achieve the plastic bag-free status is ____
a) Maharashtra b) Sikkim c) Goa d) Gujarat
- 8 In 2018, Indian government has announced the ban of single use of plastic items by ____ year.
a) 2025 b) 2022 c) 2024 d) 2000
- 9 _____ forms the largest source of ocean micro plastics worldwide as per city research, World Atlas 2018
a) Synthetic textiles b) Car tyres c) City dust d) Marine coatings
- 10 Increase in concentrations of contaminants through transfer within a food web can be termed as ____
a) Eutrophication b) Bio magnification
c) Bio stimulation d) Bio remediation
- 11 Styrene monomers are found in ____ types of plastic
a) LDPE b) HDPE c) PS d) PVC
- 12 _____ process is a part of mechanical recycling of plastics.
a) Segregation b) Pelletization c) Extrusion d) All of these

Q2. Solve the following.

- A) State the types of Plastic and explain any one. 6
- B) State the types of Recycling of Plastic Waste and explain any one in detail. 6

- Q3. Solve the following.
- A) Explain the utilisation of Plastic Waste in Road Construction. 6
 - B) Explain the problems with banning plastic. 6
- Q4. Solve Any Two of the following.
- A) Define following terms Plastic Waste (Management and Handling) Rules, 2016: 1) Compostable plastic 2) Disintegration 3) Extended producer's responsibility 4) Institutional waste generator 5) Recycling 6) Producers 6
 - B) State the Plastic Waste Management (Amendment) Rules, 2018. 6
 - C) State the types of Manufacturing process of Plastic and explain any one. 6
- Q5. Solve Any Two of the following.
- A) State and explain methods of Plastic Recycling. 6
 - B) State the benefits of Plastic to the human health and environment. 6
 - C) Explain Biodegradation or Biotic Degradation of Plastic Materials. 6
- Q6. Solve Any Two of the following.
- A) Discuss the significance of Circular Economy. 6
 - B) Define the following terms: 1) Aerobic decomposition 2) Anaerobic decomposition 3) Biological decomposition 4) Biomass 5) Bioplastics 6) Composting 6
 - C) State the Key Sources of Plastic debris. 6

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